

Practise Questions

1. Employee Management System

- **Scenario:** Each `Employee` object contains an employee's name and salary. The salary is stored dynamically to prevent unintentional data sharing during object copying.
- **Tasks:**
 1. Define a `string` member for the name and a `double*` pointer for salary.
 2. Implement shallow and deep copy constructors.
 3. Create methods to display employee details and properly manage memory in the destructor.

2. Course Enrollment System

- **Scenario:** A `Course` object stores a course name and an array of dynamically allocated grades for multiple students.
- **Tasks:**
 1. Define `string` for course name and `double*` for grades.
 2. Implement constructors to initialize these values.
 3. Write a deep copy constructor to handle dynamic memory properly.
 4. Add methods to display course details and grades, and implement a destructor to free memory.

3. Product Inventory System

- **Scenario:** Each `Product` object contains the product's name and a dynamically allocated price.
- **Tasks:**
 1. Define a `string` member for the product name and a `double*` pointer for price.
 2. Implement shallow and deep copy constructors.
 3. Add methods to update the product price, display product details, and manage memory using a destructor.

Beaconhouse National University (BNU) – SCIT Department

4. Dynamic String Manager

- **Scenario:** A `StringManager` class stores a dynamically allocated string and manages operations like copying and modifying the string.
- **Tasks:**
 1. Define a `char*` member for the string.
 2. Implement constructors for initialization.
 3. Write shallow and deep copy constructors to handle string data correctly.
 4. Include methods for string concatenation, display, and proper memory deallocation.